

XV6 Lab Execution Report & Verification

Repository: <https://github.com/Laibaqi/xv6-labs-2026.git> | Branch: problemSet1 | Environment: Ubuntu WSL2 RISC-V QEMU

1. sleep Utility

```
[Laiba Irfan - 31736 @ xv6-labs-2026]
Testing sleep ticks system call in XV6 shell:
$ sleep 5
$ echo done
done
```

2. pingpong Utility

```
[Laiba Irfan - 31736 @ xv6-labs-2026]
Testing two-way pipe process communication:
$ pingpong
4: received ping
3: received pong
```

3. primes Utility

```
[Laiba Irfan - 31736 @ xv6-labs-2026]
Testing concurrent prime sieve pipeline using pipes:
$ primes
prime 2
prime 3
prime 5
prime 7
prime 11
prime 13
prime 17
prime 19
prime 23
prime 29
prime 31
```

4. find Utility (-exec support)

```
[Laiba Irfan - 31736 @ xv6-labs-2026]
Testing recursive directory traversal & command execution:
$ find . README
./README
$ find . findtest.sh -exec grep hello
hello
hello
hello
hello
hello
```

5. xargs Utility

```
[Laiba Irfan - 31736 @ xv6-labs-2026]
Testing xargs command line argument piping:
$ echo hello | xargs echo
hello
```

6. sixfive Utility

```
[Laiba Irfan - 31736 @ xv6-labs-2026]
Testing number separator filtering:
$ sixfive sixfive.txt
5
100
18
6
```

7. memdump Utility

```
[Laiba Irfan - 31736 @ xv6-labs-2026]
Testing memory dump formatting:
$ memdump
Example 1:
61810
2026
Example 2:
```

```
a string
Example 3:
another
Example 4:
hello
1819438967
100
z
xyzyy
```

8. uptime Utility

```
[Laiba Irfan - 31736 @ xv6-labs-2026]
Testing uptime tick counter:
$ uptime
105
```